S.C.O.U.T.

Santa Clara Oceanic Utilities Transmitter

MVP System Overview & Framework

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https://github.com/irodriguez-17/SCOUT

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Project Vision

S.C.O.U.T. is a **small, solar-powered, modular marine buoy system** for **long-term shallow-water coral reef monitoring**. Deployed adjacent to reefs, it collects environmental data indicative of reef health.

Design priorities:

- Affordable
- Durable
- Low power
- Scalable (many buoys, many locations globally)
- Modular (future sensing applications)
- Minimal maintenance
- Small, visually simple form factor

Long-term vision: A global reef monitoring network with many SCOUT buoys deployed across reef systems.

Primary Mission (MVP)

Deploy a **solar-powered buoy adjacent to coral reefs** that:

- Records reef-relevant environmental data
- Operates autonomously for **1+ year deployments**
- Conserves energy via sleep/wake cycling
- Wirelessly transmits data to a **shore-side base station**
- Enables reef-health monitoring and statistical analysis

MVP Data Collection

1. Temperature Sensor (Required)

Purpose: Monitor reef stress and bleaching risk.
Why: Coral bleaching strongly correlates with prolonged elevated temperature.
Sampling: ~Every **30–60 minutes** (very low power).

2. Water Quality Proxy (Required - At Least One)

Option A: Turbidity

- Measures suspended particles / water clarity
- Indicates sedimentation, runoff, storm disturbance, poor reef conditions
- Suggested sampling: **1–4 times/day**

Option B: Light

- Measures photosynthetically available light
- Important for coral symbiotic algae health
- Suggested sampling: **1–4 times/day**

Current recommendation: Temperature + Turbidity, with light added if feasible.

3. Hydrophone / Acoustic Monitoring (Optional / Stretch Goal)

Purpose: Record reef soundscapes.
Why: Reef acoustic signatures vary with biodiversity, reef health, and stress.

Approach:

- Short recordings at periodic intervals
- Shore-side filtering removes:
    - Boat noise
    - Wave noise
    - Environmental interference

Potential value:

- Baseline reef sound signatures
- Long-term acoustic health trends

Status: High value, but higher scope and complexity.

Power System

Design Philosophy

Ultra-low-power operation through intelligent duty cycling.

Operating logic:

- Sleep most of the time
- Wake to record data
- Store locally
- Return to sleep/charge
- Periodically transmit to shore

Energy Resilience Strategy (Cloudy Periods)

System priorities:

1. Maintain core operation
2. Reduce transmission frequency
3. Continue local data storage
4. Pause nonessential sensing if battery becomes critical
5. Resume normal operation after recharge

Proposed Components

- Solar panel
- Rechargeable battery
- Charge controller
- Power monitoring system
- Low-power embedded controller

Likely battery: LiFePO4 (Lithium Iron Phosphate) for lifespan, durability, safety, and common marine use.

Embedded Electronics & Control

Proposed Control Logic

Sleep → Wake → Record → Store → Battery Check → Transmit if power sufficient → Sleep

If low power:

- Delay transmission
- Continue energy conservation

Hardware Direction

ESP32-based system (current recommendation)

Why:

- Strong low-power sleep modes
- More capable than standard Arduino
- Well suited for IoT sensing
- Modular sensor integration

Responsibilities:

- Sensor management
- Sleep scheduling
- Battery monitoring
- Power optimization
- Local data logging
- Radio communication

Communications System

Primary Method:

LoRa Radio

Why:

- Very low power
- Long range
- Reliable for small environmental data packets
- Well-suited for autonomous sensing systems

Current assumptions:

- Max range: **~100 yards**
- Direct line of sight to shore
- No dependence on internet or cellular

Key concern: Saltwater RF interference requires testing/validation.

Mechanical / Marine Design

Deployment Conditions

- Fixed mooring
- **5–8 meter max depth**
- Buoy remains at surface

Geometry

Shape:

- Cylindrical at waterline
- Tapers downward to mooring attachment
- Tapers upward toward antenna

Purpose:

- Stability
- Reduced drag
- Clean/elegant appearance
- Functional solar + antenna integration

Sensor Placement

- Mounted beneath buoy **or**
- Suspended lower on anchor/mooring chain

Placement determined by:

- Data quality
- Depth performance
- Reliability

Requirements

Must be:

- Waterproof
- Storm resistant
- Biofouling resistant
- Serviceable
- Reef-safe

Anchoring: Long-term stability with minimal reef disturbance.

Shore-Side Base Station

Purpose

Receive and process buoy data.

Likely hardware: Raspberry Pi receiver system.

Responsibilities:

- LoRa reception
- Local storage
- Basic statistical processing
- Quantitative sorting/organization

Future additions:

- Cloud upload
- Dashboards
- ML/data analysis
- Multi-buoy network management

Overall System Architecture

1. **Environmental Sensing:** Temperature, turbidity/light, optional hydrophone
2. **Embedded Control:** Sleep/wake, sensing, battery management, storage
3. **Power:** Solar, battery, charge control, optimization
4. **Communications:** LoRa buoy-to-shore transmission
5. **Marine Housing:** Waterproof buoy, mooring, anti-fouling, storm survivability
6. **Shore Station:** Raspberry Pi, storage, analytics, future cloud/ML

Long-Term Vision

A **globally scalable reef monitoring platform** supporting:

- Coral reef conservation

- Long-term environmental monitoring
- Reef restoration
- Researchers
- NGOs
- Resorts/coastal stakeholders
- Governments

Goal: Make reef monitoring affordable, continuous, modular, and globally scalable.